

PERSONAL HEALTHCARE MONITORING SYSTEM

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INTRODUCTION

In today's fast-paced lifestyle, maintaining good health has become very important. Regular monitoring of physical parameters such as Body Mass Index (BMI), sleep hours, and daily physical activity helps in identifying potential health risks at an early stage.

This project, **Personal Health Monitoring System**, is developed using Python programming language. The system collects user inputs like weight, height, sleep hours, and daily steps. Based on these inputs, it calculates BMI, generates a health score, and provides basic health recommendations.

The system is simple, user-friendly, and runs in a command-line interface.

OBJECTIVE

The main objectives of this project are:

- To calculate Body Mass Index (BMI)
- To evaluate overall health score
- To analyze sleep and physical activity
- To detect possible health risks
- To store user health records using JSON file
- To provide basic diet and exercise suggestions

SYSTEM REQUIREMENTS

Hardware Requirements

- Computer / Laptop

- Minimum 4GB RAM

Software Requirements

- Python 3.x
- Command Prompt / Terminal
- Text Editor (VS Code / Notepad)

METHODOLOGY

The system follows the below steps:

Step 1: Input Collection

The user enters:

- Name
- Age
- Weight (kg)
- Height (cm)
- Daily steps
- Sleep hours
- Health goal

Step 2: BMI Calculation

BMI is calculated using the formula:

$$BMI = \frac{Weight(kg)}{Height(m)^2}$$

Height is converted from centimeters to meters before calculation.

Step 3: Health Score Calculation

Health score starts from 100.

Marks are deducted based on:

- Abnormal BMI
- Insufficient sleep

- Low physical activity

Step 4: Recommendation Generation

Using conditional statements:

- Nutrition plan is generated
- Exercise recommendation is given
- Sleep analysis is provided

Step 5: Risk Detection

The system checks for:

- Obesity risk
- Malnutrition risk
- Sleep disorder risk

Step 6: Data Storage

All user data is stored in a JSON file (health_data.json) for future reference.

SYSTEM DESIGN

Functional Modules

1. BMI Calculation Module
2. Health Score Module
3. Recommendation Module
4. Risk Detection Module
5. Data Storage Module

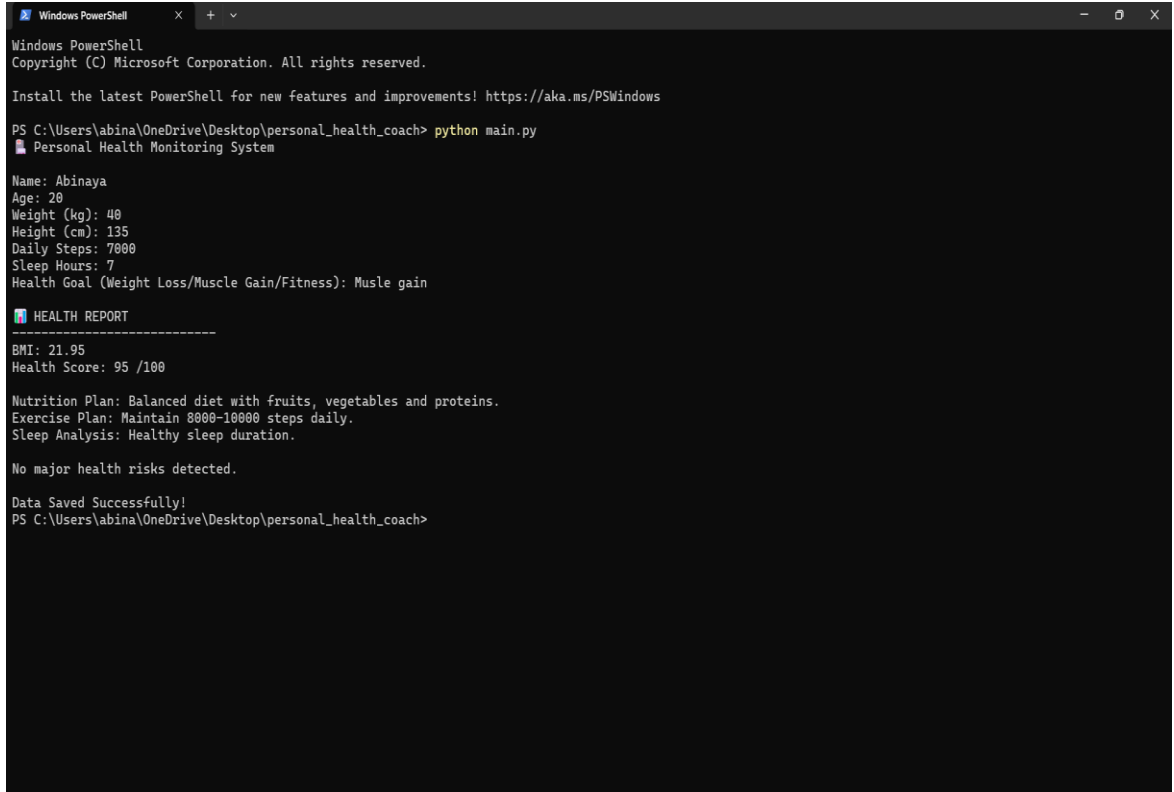
The system uses Python functions to implement each module.

OUTPUT

The system displays:

- BMI value
- Health Score (Out of 100)
- Nutrition Plan

- Exercise Plan
- Sleep Analysis
- Risk Alerts (if any)
- Confirmation of Data Saved



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Windows PowerShell
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PS C:\Users\abina\OneDrive\Desktop\personal_health_coach> python main.py
Personal Health Monitoring System

Name: Abinaya
Age: 20
Weight (kg): 40
Height (cm): 135
Daily Steps: 7000
Sleep Hours: 7
Health Goal (Weight Loss/Muscle Gain/Fitness): Muscle gain

HEALTH REPORT
-----
BMI: 21.95
Health Score: 95 /100

Nutrition Plan: Balanced diet with fruits, vegetables and proteins.
Exercise Plan: Maintain 8000-10000 steps daily.
Sleep Analysis: Healthy sleep duration.

No major health risks detected.

Data Saved Successfully!
PS C:\Users\abina\OneDrive\Desktop\personal_health_coach>

```

ADVANTAGES

- Simple and user-friendly
- Easy to implement
- Low system requirements
- Helps monitor personal health
- Stores data for tracking progress

LIMITATIONS

- Command-line based interface
- Basic health recommendations only
- Does not replace professional medical advice

FUTURE ENHANCEMENT

- Add graphical user interface (GUI)
- Add graphical health progress charts
- Add email report system
- Add database integration
- Develop mobile application version

CONCLUSION

The Personal Health Monitoring System successfully calculates BMI and evaluates overall health using various parameters such as sleep and physical activity. The system provides useful health recommendations and stores user data efficiently using JSON format.

This project demonstrates the use of Python programming concepts such as functions, conditional statements, file handling, and data storage.